

Implementation of a QPSK System on FPGA

S.O. Popescu*, A.S.Gontean* and D.Ianchis*

*Applied Electronics Departament, Faculty of Electronics and Telecommunications,
Politehnica University of Timisoara, Romania;

silvana.popescu@etc.upt.ro, aurel.gontean@etc.upt.ro, daniel.ianchis@etc.upt.ro

Abstract— The paper presents the design of a QPSK digital communication system and its implementation on FPGAs. The simulations of the system were made in Matlab/Simulink environment and System Generator, a tool used for FPGA designs. The system is made up of two Spartan 3E Starter Kit boards. The first is implemented to comport as a modulator and the second as a demodulator. The VHDL hardware description language was used to design each model. The local clock oscillator on the board operates at 50Mhz and used at 31,250 kHz for the carrier. The operating results were compared with the theorethical ones and were also evaluated by measurements.

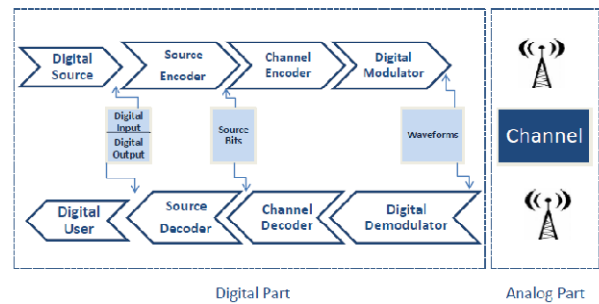


Figure 1. A Digital Communication System [8].

I. INTRODUCTION

A significant transition from analog to digital has occurred in the last years and communications is a field which has gain a lot because of this development. The advanced algorithms used in a digital communication system made it more reliable than an analog one.

Studies about the implementation of digital communication systems were made in [1]-[7].

The hardware and software resources used in generating the QPSK (Quadrature Phase Shift Keying) modulation and demodulation were a computer with the Xilinx WebPack ISE on it, two Spartan 3E Starter Kit boards and a LeCroy WaveSurfer Xs Series Oscilloscope, a high performance digital oscilloscope.

The purpose of this paper is to create a QPSK (Binary Phase Shift Keying) system made of a modulator, a channel and a demodulator from two Spartan 3E boards. The modulated signal was obtained in the first Spartan 3E board, passed through a channel and transmitted to the second board, the demodulator. At the output of the demodulator, the modulating signal was found again.

The paper is organized into 6 sections. The paper begins with an introduction in section 1. Section 2 presents the theoretical backgrounds about the digital communication system and about the QPSK modulation and demodulation. The implementation of the QPSK system in Matlab/ Simulink and System Generator are presented in section 3. Section 4 is dedicated to the implementation of the system: modulator and demodulator on the Spartan 3E Starter Kit boards. The results are discussed in section 5. The final section, 6, presents the conclusions.

II. THEORETICAL BACKGROUNDS

A. Digital Communication System

A typically digital communication system is presented in Fig.1. The role of a digital communication system is to transport digital data between two nodes: the transmitter and the receiver. A digital communication system is made up of both digital and analog parts. The digital part

consists of digital source/user, source encoder/decoder, channel encoder/decoder and the digital modulator/demodulator. The transmitter, receiver, the channel models and noise models forms the analog part [1].

The message to be sent is from a digital source. The binary information obtained at the output of the channel encoder is than passed to a digital modulator with the main purpose to translate the discrete symbols into an analog waveform that can be transmitted over the channel [9], [10]. In the receiver, the reverse signal processing happens. The channel represents the physical medium that carries a signal between the transmitter and the receiver. Any signal which passes through the channel is corrupted with noise whatever the medium used for transmission [8].

The digital data is transmitted between the transmitter and the receiver by varying a physical characteristic of a sinusoidal carrier, either the frequency or the phase or the amplitude. This operation is performed with a modulator at the transmitting end to impose the physical change to the carrier and a demodulator at the receiving end to detect the resultant modulation on reception [2], [8].

B. QPSK Modulation and Demodulation

The signals of the QPSK modulation are defined in (1), where f_c is the frequency of the carrier and $0 \leq t \leq T$ [8], [9]:

$$s_i(t) = A \cos[2\pi f_c t + \theta_i], \quad i = 1, 2, 3, 4 \quad (1)$$

$$\text{and } \theta_i = \frac{(2i-1)\pi}{4}$$

The initial phases of the signals are $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$.

Equation (1) can also be written as [8], [9]:

$$s_i(t) = s_{i1}(t)\phi_1(t) + s_{i2}(t)\phi_2(t) \quad (2)$$

where:

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos 2\pi f_c t, 0 \leq t \leq T \quad (3)$$

$$\phi_2(t) = -\sqrt{\frac{2}{T}} \sin 2\pi f_c t, 0 \leq t \leq T \quad (4)$$

$$s_{i1} = \sqrt{E} \cos \theta_i \quad (5)$$

$$s_{i2} = \sqrt{E} \sin \theta_i \quad (6)$$

where $E = \frac{A^2 T}{2}$ is the symbol energy.

In QPSK, two bits are grouped together to form signals, and one of the four possible cases of dibits: 00, 01, 10 and 11 is transmitted. Also, the odd bits were mapped to s_{i1} and the even bits into s_{i2} [2].

Equation (2) of the entire QPSK signal can be written as:

$$s(t) = \frac{A}{\sqrt{2}} I(t) \cos 2\pi f_c t - \frac{A}{\sqrt{2}} Q(t) \sin 2\pi f_c t \quad (7)$$

where $I(t)$ and $Q(t)$ are pulse trains determined by the odd and even bits [9].

Since each bit occupies T_b seconds, the signals corresponding to the dibits last for $T_s = 2T_b$ [9].

The QPSK modulator illustrated in fig.2 is based on (7). The binary sequence is separated by the serial-to-parallel converter into odd-bit-sequence for I channel and even-bit-sequence for Q channel. The corresponding waveforms are illustrated in fig.3.

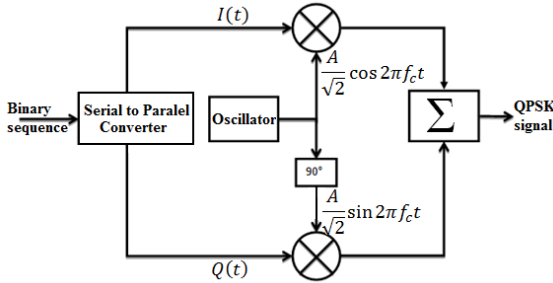


Figure 2. The QPSK Modulator [9].

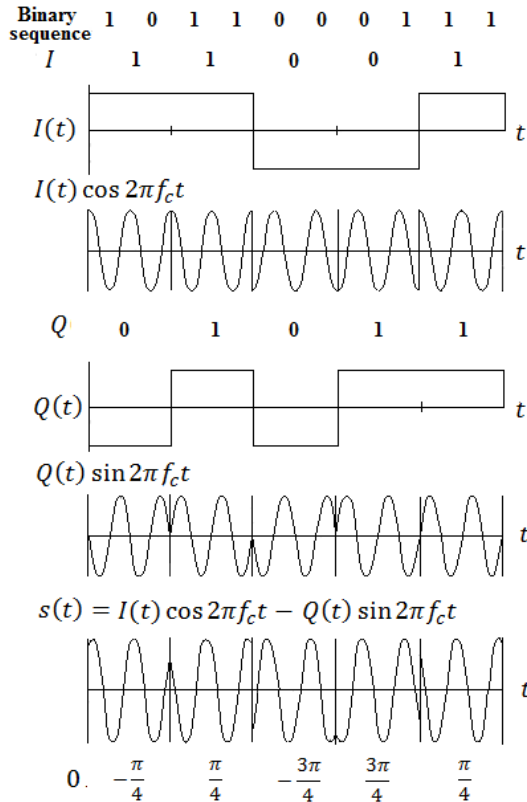


Figure 3. QPSK Waveforms [9].

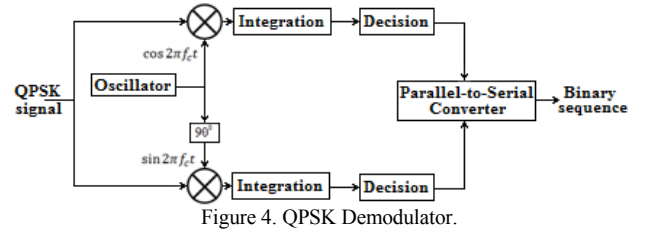


Figure 4. QPSK Demodulator.

In the demodulator, the reverse operation happens. The I-channel and the Q-channel are demodulated separately as two different BPSK signals. The I-channel is demodulated with a cosine waveform and the Q-channel, with a sine waveform. A parallel-to-serial-converter is used to combine the two sequences into a single one which represents the modulating signal.

III. QPSK SYSTEM

A. QPSK System in Simulink

Fig. 5 represents a communication system implemented in the Matlab/ Simulink environment that uses the BPSK modulation technique. The system is composed of the binary data source, a modulator, a channel and a demodulator.

In [10], a QPSK modulator is implemented in the Simulink environment and in System Generator. We used the implementation of the modulator [10] in order to create the system.

The modulator [10]: The binary data source is made of a random data source and a rounding function and generates the modulating signal. The binary sequence is separated into an odd-bit-sequence I and an even-bit-sequence with the help of the sample-and-hold block. The I-channel is modulated with a cosine waveform generated by a cosine block and the Q-channel is modulated with a sine waveform. The switch blocks will choose between the first or third output depending on the value of the second input. In the case of the I-channel, if the second input is "1", the output value will be cosine, but if the second input is "0", the output will be $-\cos$ and for the Q-channel, if the second input is "1", the output value will be sine, but if the second input is "0", the output will be $-\sin$. The sum block combines the two modulated signals into the QPSK signal.

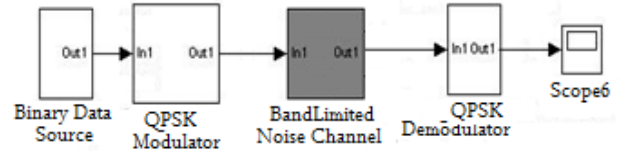


Figure 5. QPSK System.

The corresponding signal in the modulator can be seen in fig.6.

The modulated signal is then pass through a channel where noise is added (fig.7). The waveform of the modulated signal with noise is shown in fig.8.

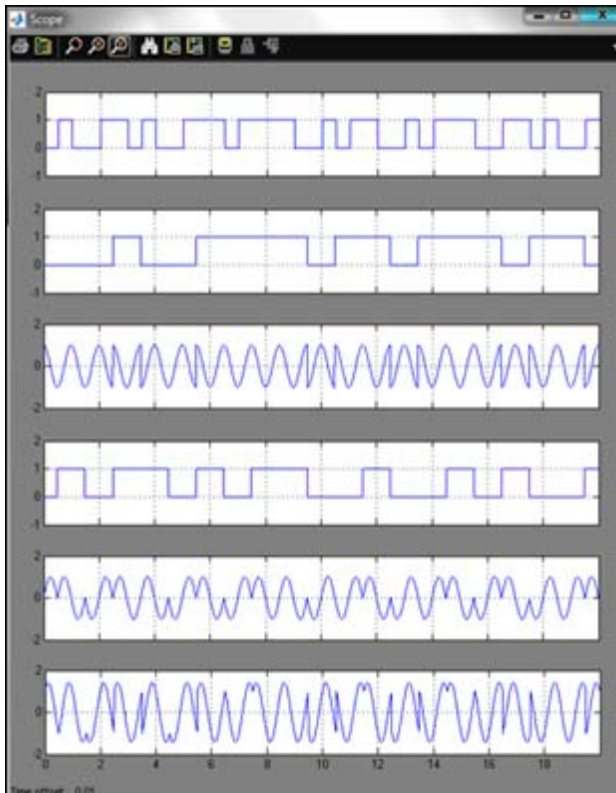


Figure 6. The waveforms in the modulator [10]
 (a) The modulating signal (b) $I(t)$ (c) $I(t) \cos 2\pi f_c t$
 (d) $Q(t)$ (e) $Q(t) \sin 2\pi f_c t$ (f) The modulated signal

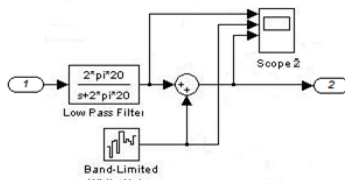


Figure 7. Band Limited Noisy Channel.

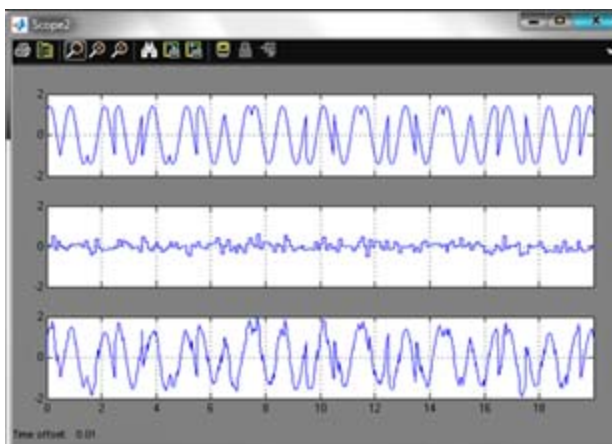


Figure 8. The waveforms shown on the scope
 (a) the modulated signal (b) the noise
 (c) the modulated signal with noise.

The modulated signal added with noise arrives at the demodulator (fig. 9) and is divided into two channels. The first channel is modulated with a cosine waveform and the second with a sine as well as in theory. At the outputs, the I and Q channels are found and put together to obtain the initial binary sequence (fig. 10).

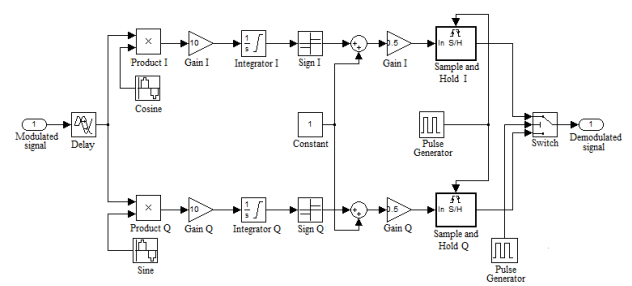


Figure 9. QPSK demodulator.

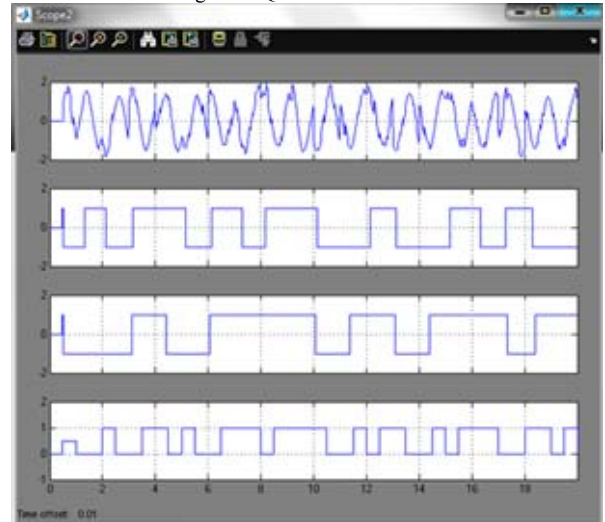


Figure 10. The waveforms in the demodulator
 (a) The modulated signal with noise (b) The I-channel
 (c) The Q-channel (d) The demodulated signal.

B. QPSK System in System Generator

Designs made in System Generator include all implementation steps: synthesis, place and route in order to generate an FPGA programming file [11]. The main advantage of System Generator is making possible the administration of the Xilinx components.

Our QPSK system implemented in System Generator uses the same modulator implemented in [10], the data source, a channel and a demodulator.

The modulating signal is generated internal by a LFSR (Linear Feedback Shift Register). The Time Division Demultiplexer block divides the modulating signal into the odd-sequence I and even-sequence Q, action which takes place inside the FPGA (fig.11). Then, the I-channel is modulated with cosine and the Q-channel with sine (fig.12). The mux block implements a multiplexer. It has 3 inputs: the d0 and d1 inputs of the mux represent cosine and $-\cos$ for the I sequence and sine and $-\sin$ for the Q sequence. The sel input selects between d0 and d1 depending on the odd-sequence for mux1 and the even-sequence for mux. The modulated signal is obtained by adding the two outputs. The modulated signal is then pass through the same channel used in Simulink where noise is added and arrives at the input of the demodulator (fig.13a).

In the demodulator, the modulated signal with noise is multiplied once with sine, and the second time with cosine. The sine and cosine waveforms are recovered in Simulink with the sine and cosine blocks.

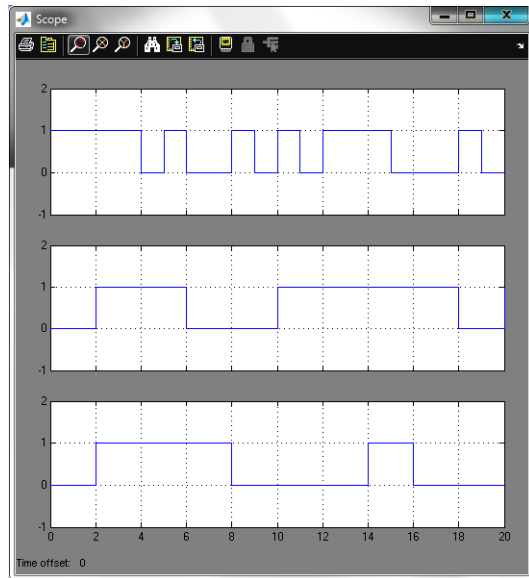


Figure 11. The waveforms in the modulator:
(a) The modulating signal (b) The I-channel
(c) The Q-channel

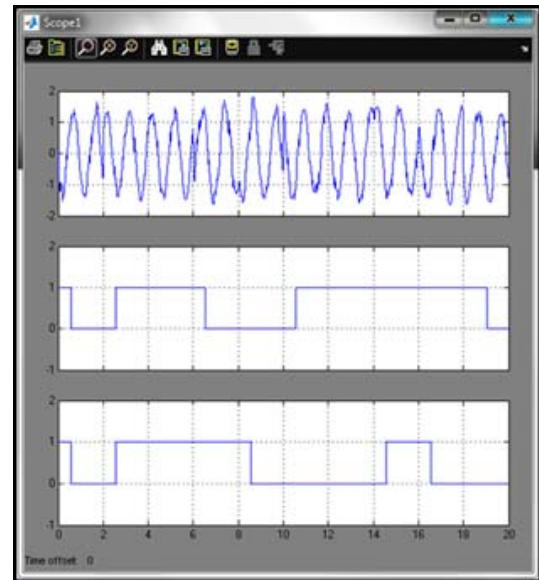


Figure 14. The Waveforms:
(a) the modulated signal with noise (b) the demodulated I-channel
(c) the demodulated Q-channel.

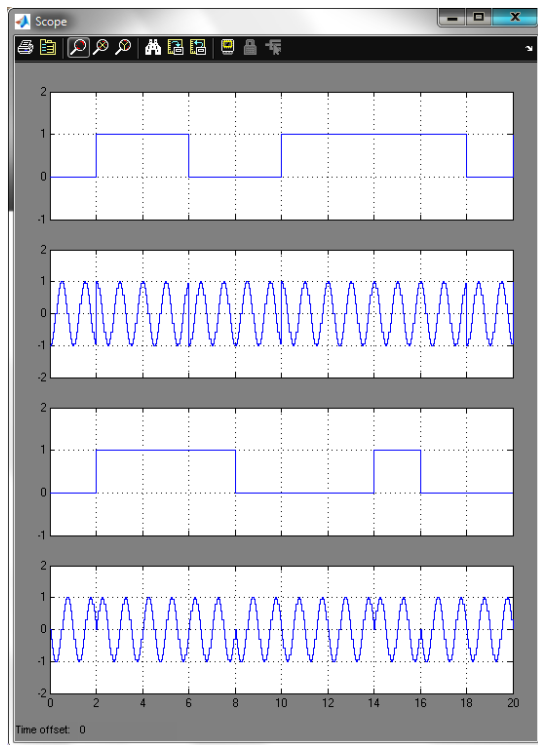


Figure 12. The Waveforms:
(a) I-channel (b) Q-channel (c) the modulated signal.

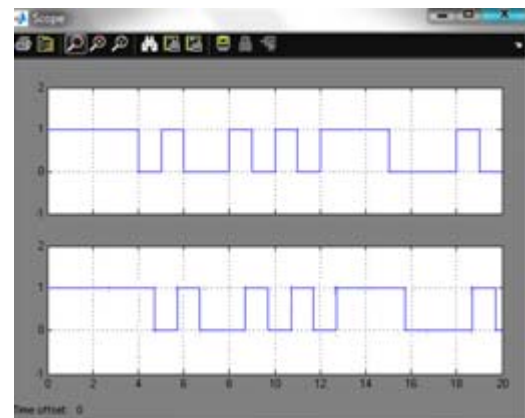


Figure 15. The Waveforms:
(a) the modulating signal (b) the demodulated signal.

The two signals obtained are kept in two accumulators and then compared with a threshold. If the compared signal is positive, the demodulator takes the decision that '1' was transmitted, otherwise, '0', obtaining the demodulated I and Q channels (fig.14b and fig.14c). The multiplexer is the one who puts the I and Q channels together obtaining the demodulated signal which is the initial modulating signal (fig.15).

IV. BPSK SYSTEM ON THE SPARTAN 3E BOARD

The BPSK System (fig.16) that we implemented on the Spartan 3E Starter Kit board [12] has as a model the implementation in System Generator with the only difference that the sine and cosine signals are generated internal, in a ROM.

The BPSK system (fig.16) consists of two Spartan 3E boards, the first board behaves as a modulator and the second one, as a demodulator. The connections between the two boards are made of three wires: the green wire comports as a communication channel, the red wire represents the synchronization between the two boards and the black one comports as an asynchronous reset signal.

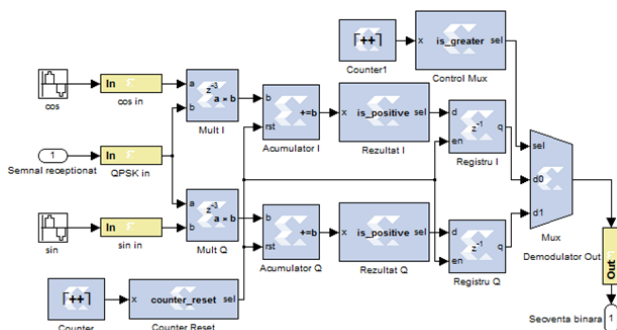


Figure 13. QPSK Demodulator in System Generator.

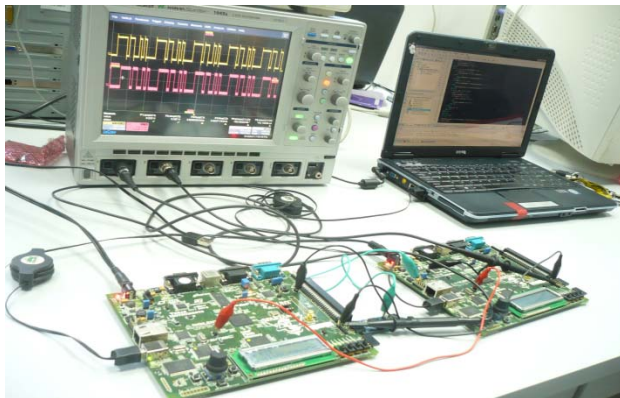


Figure 16. QPSK System – experimental setup.

The modulating signal is generated internal, in the modulator, by a LFSR and is divided into two sequences: the-odd-sequence or the I-channel and the-even-sequence or the Q-channel. The I-channel is modulated with cosine and the Q-channel with sine. The sine and cosine are also generated internal, and are made of 16 different values kept in a ROM memory [13]. The cosine is obtained from sine by reading the values later with 4 samples. By adding the two modulated signals, the QPSK modulated signal is obtained. The modulated signal is then sent to the DAC (Digital-to-Analog Converter) on the board in order to be sent through the channel. The principle of the QPSK modulator implemented on the FPGA is shown in fig.17.

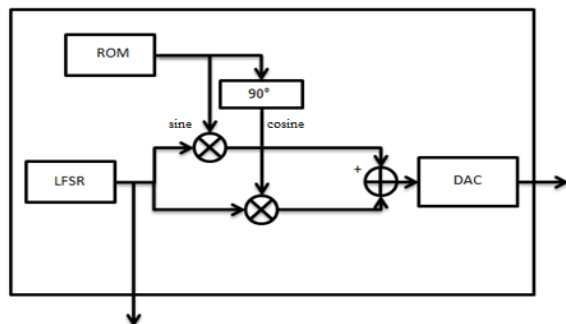


Figure 17. The principle of the QPSK modulator on the FPGA [10].

The modulated signal affected with noise arrives at the second board which behaves as a demodulator. The signal is converted into a digital form with the help of a pmod AD1. This digital signal is then multiply with both signals, the sine and cosine generated internal in a ROM memory. The results are kept in an accumulator and compared with a decision threshold and so, the demodulated I-channel and Q-channel are obtained and in the end the demodulated signal. The principle of the QPSK demodulator implemented on the FPGA is illustrated in fig.18.

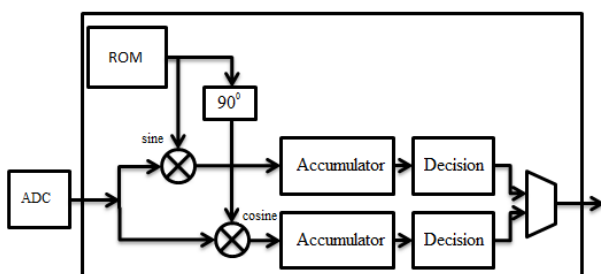


Figure 18. The principle of the QPSK demodulator on the FPGA

V. RESULTS

After implementing the QPSK System made of a modulator and demodulator on the two Spartan 3E Starter Kit boards, the signals were routed to the LeCroy WaveSurfer Xs Series Oscilloscope [14], a high performance digital oscilloscope.

Fig.19 illustrates two signals in the modulator: the LFSR and the QPSK modulated signal. In fig.20, the modulating and the demodulated signals are shown.

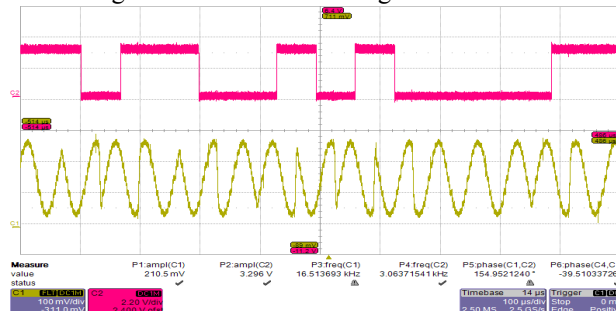


Figure 19. The Waveforms:

(a) the modulating signal (LFSR) ; (b) the modulated signal.

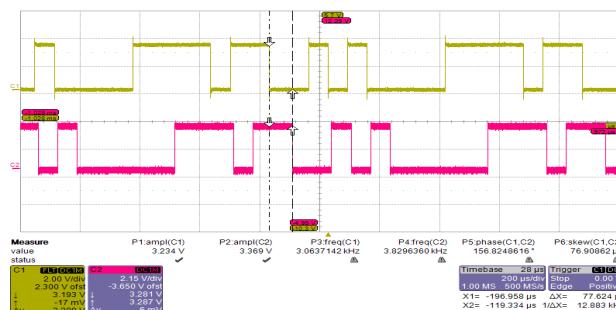


Figure 20. The waveforms:

(a) The modulating signal (b) The demodulated signal.

Fig. 21 and 22 illustrate the design summary of the modulator and demodulator boards.

Device Utilization Summary				
Logic Utilization	Used	Available	Utilization	Note(s)
Number of Slice Flip Flops	95	9,312	1%	
Number of 4 input LUTs	157	9,312	1%	
Number of occupied Slices	113	4,656	2%	
Number of Slices containing only related logic	113	113	100%	
Number of Slices containing unrelated logic	0	113	0%	
Total Number of 4 input LUTs	164	9,312	1%	
Number used as logic	157			
Number used as a route-thru	7			
Number of bonded IOBs	14	232	6%	
Number of BUFMUXs	2	24	8%	
Average Fanout of Non-Clock Nets	3.29			

Figure 21. Design Summary of the QPSK Modulator.

Device Utilization Summary				
Logic Utilization	Used	Available	Utilization	Note(s)
Number of Slice Flip Flops	233	9,312	2%	
Number of 4 input LUTs	277	9,312	2%	
Number of occupied Slices	221	4,656	4%	
Number of Slices containing only related logic	221	221	100%	
Number of Slices containing unrelated logic	0	221	0%	
Total Number of 4 input LUTs	304	9,312	3%	
Number used as logic	277			
Number used as a route-thru	27			
Number of bonded IOBs	15	232	6%	
Number of BUFMUXs	3	24	12%	
Number of MULT18X18SIOs	2	20	10%	
Average Fanout of Non-Clock Nets	2.62			

Figure 22. Design Summary of the QPSK Demodulator.

Fig.23 and fig.24 show the post and route implementation of the QPSK modulator and demodulator on the two Spartan 3E boards.

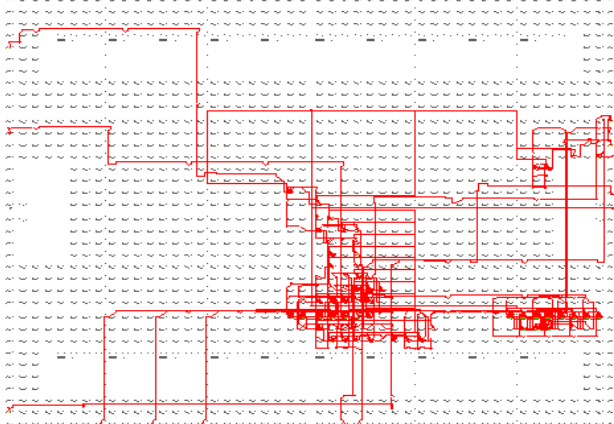


Figure 23. The route paths of the QPSK modulator.

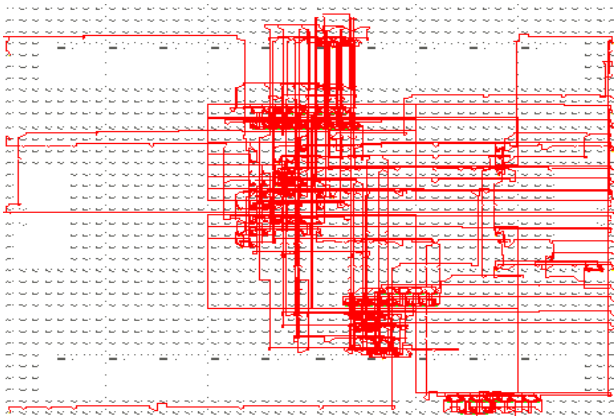


Figure 24. The route paths of the QPSK demodulator.

VI. CONCLUSION

The paper presents an implementation of a QPSK System (Modulator and Demodulator) in the Matlab/Simulink environment as well as a proposal of a QPSK System in System Generator.

The binary sequence is generated internal by a LFSR and is divided into two sequences: the odd-sequence (I-channel) and the even-sequence (Q-channel). The odd-sequence is modulated with the cosine waveform and the even-sequence with the sine waveform, waveforms obtained outside the board. The QPSK modulated signal is obtained by adding the two modulated signals. The QPSK modulated signal is then passed through the same channel used in Simulink where noise is added. In the demodulator, the modulated signal with noise is multiplied twice, once with sine, and the second time with cosine. The two signals are kept in two accumulators and then compared with a threshold. If the compared signal is greater than the threshold, the demodulator takes the decision that '1' was transmitted, otherwise, '0', obtaining the demodulated I and Q channels. The multiplexer is the one who puts the I and Q channels together obtaining the demodulated signal which is the initial modulating signal.

The same principle is applied in implementing the modulator and the demodulator on the two boards, with the difference that the sine and cosine signals were

generated inside the board. The sine signal was made of 16 different values kept in a ROM memory and the cosine was obtained reading the ROM memory later with 4 samples. The odd-sequence was modulated with the cosine waveform and the even-sequence with the sine waveform. Adding the two modulated signals, the QPSK signal was obtained. The modulated signal is then sent to the DAC on the board in order to be sent through the channel. The modulated signal affected with noise arrives at the demodulator. The signal is converted into a digital form with the help of an AD board. This digital signal is then multiply with both signals, the sine and cosine generated internal in the ROM memory from the second board. The results are kept in an accumulator and compared with a decision threshold and so, the demodulated I-channel and Q-channel are obtained and in the end the demodulated signal.

ACKNOWLEDGMENT

This work was partially supported by the strategic grant POSDRU 6/1.5/S/13, (2008) of the Ministry of Labour, Family and Social Protection, Romania, co-financed by the European Social Fund – Investing in People.

REFERENCES

- [1] A.Gürgör, F.Arikan, O.Arikan, "Simulation of a digital communication system", in *Proceedings of the 13th European Signal Processing Conference EUSIPCO*, 2005, Turkey, <http://www.eurasip.org/Proceedings/Eusipco/Eusipco2005/deferred/papers/cr1142.pdf>
- [2] Y.Tachwali, H.Refaei, "Implementation of a BPSK Transceiver on Hybrid Software Defined Radio Platforms", 3rd International Conference on Information and Communication Technologies: From Theory to Applications, Syria, 2008, pp. 1-5.
- [3] P.Dondon, J.M.Micouleau, J.Legall, .K.Kadionik, "Design of a low cost BPSK modulator/demodulator for a practical teaching of digital modulation techniques", in the 4th WSEAS/IASME International Conference on Engineering Education, Greece, 2007, pp.61-66.
- [4] W.Song, J.Zhang, Q.Yao, "Design and Implementation of BPSK Modulator and Demodulator on Modern DSP Technology", 3rd IEEE International Symposium on Microwave, Antenna, Propagation and EMC Technologies for Wireless Communications, China, 2009, pp.1135-1137.
- [5] G.Elamary, G.Chester, J.Neasham, "A Simple Digital VHDL QPSK Modulator Designed Using CPLD/ FPGAs for Biomedical Devices Applications", in *Proceedings of the World Congress on Engineering 2009 Vol.1*, 2009, United Kingdom, pp.
- [6] P.Dondon, J.M.Micouleau, P.Kadionik, "Improving learning efficiency for digital modulation courses", WSEAS Engineering education, 2005, Greece, pp.
- [7] W.Song, Q.Yao, "Design and Implementation of QPSK Modem Based on FPGA", 2010,
- [8] S.Popescu, A.Gontean, M.Babaita, "BPSK System on Spartan 3E FPGA", *unpublished*.
- [9] F.Xiong, "Digital Modulation Techniques", *Artech House*, UK, 2000.
- [10] S.Popescu, A.Gontean, D.Ianchis, "QPSK Modulator on FPGA", *unpublished*.
- [11] System Generator for DSP. Getting Started Guide. *Xilinx*. 2008
- [12] Spartan 3E FPGA Starter Kit board. User guide. *Xilinx*. 2011.
- [13] F.Ahamed, A.Scarpino, "An educational digital communications project using FPGAs to implement a BPSK Detector", *IEEE Transactions on Education*, Vol.48, No.1, 2005, pp.191-197.
- [14] LeCroy WaveSurfer, Xs Series Oscilloscope Operator's Manual, 2008